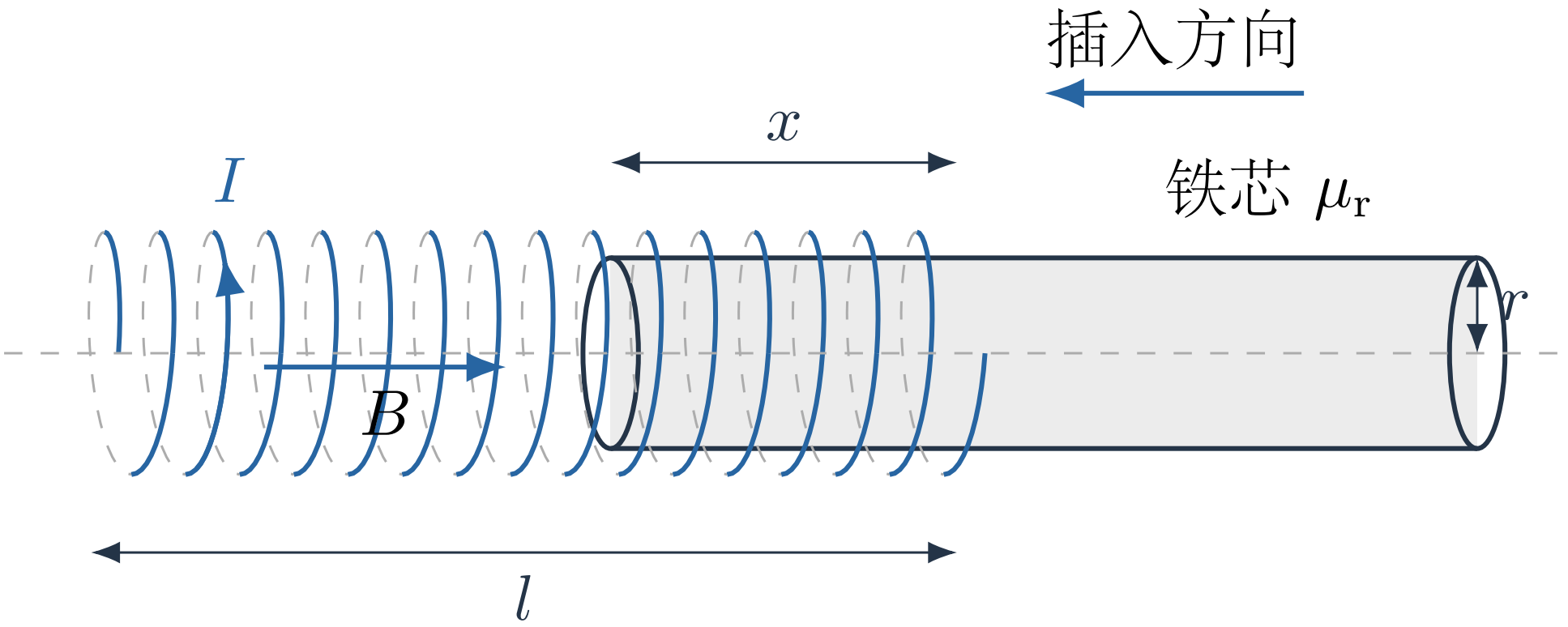
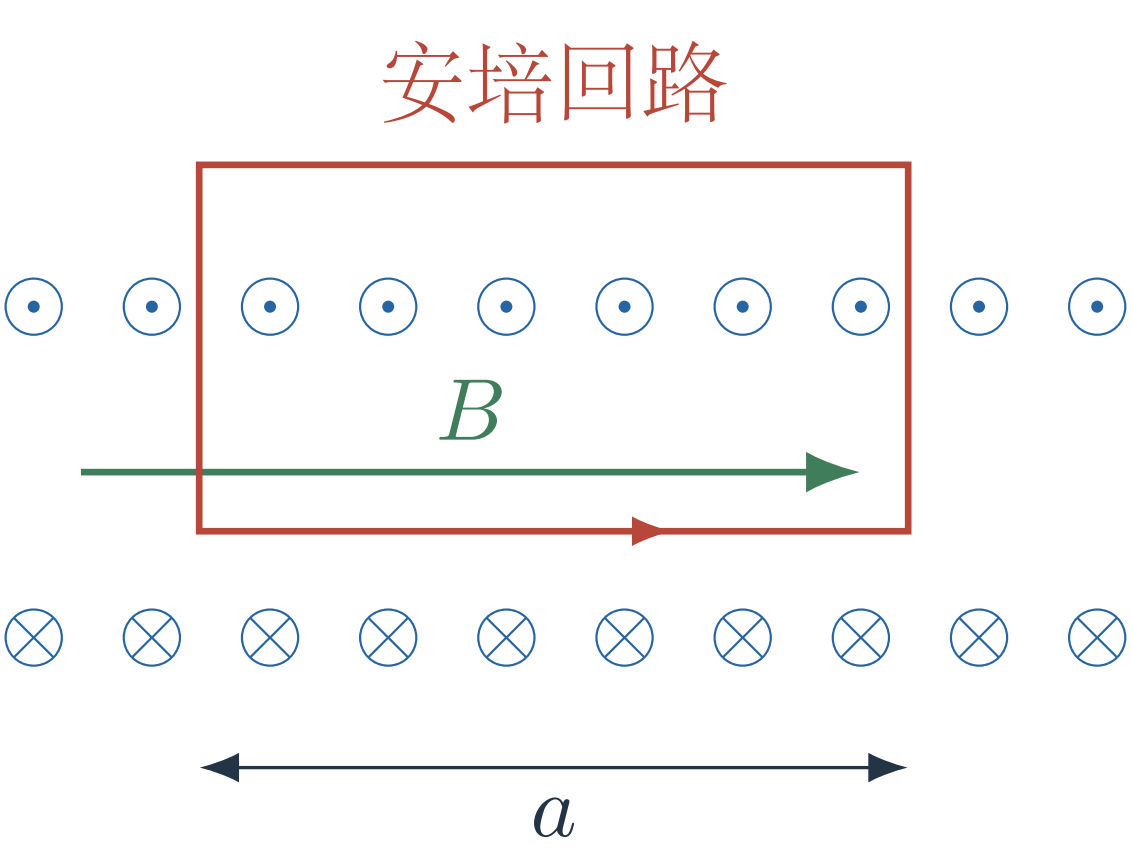
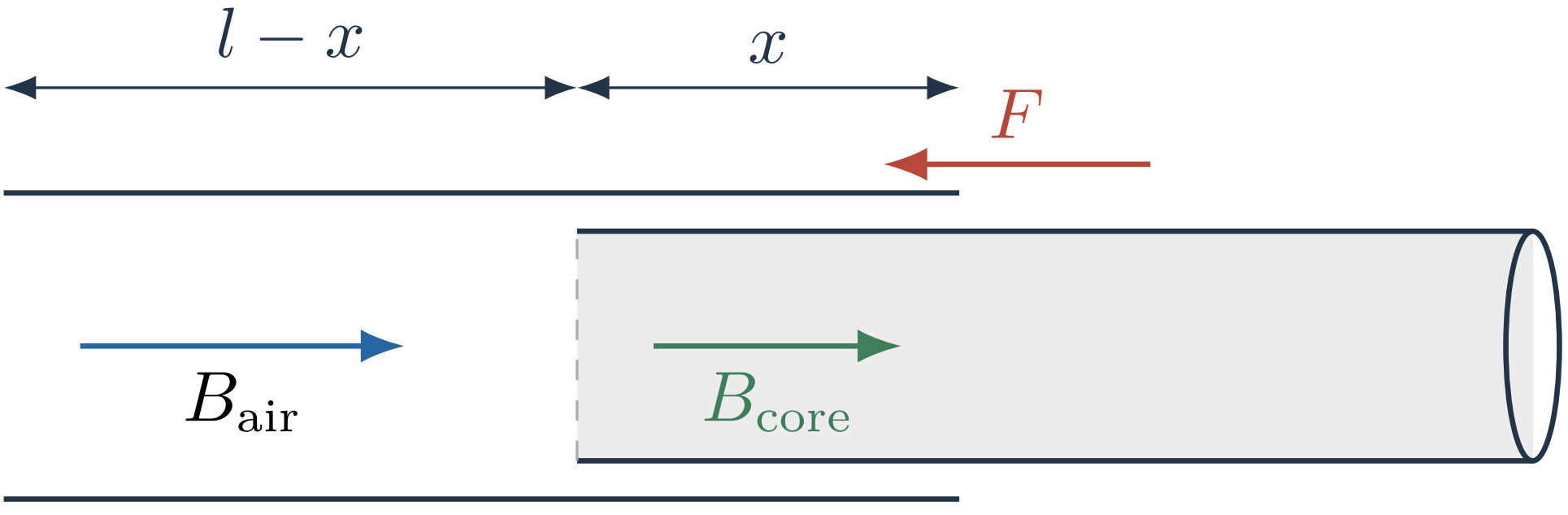
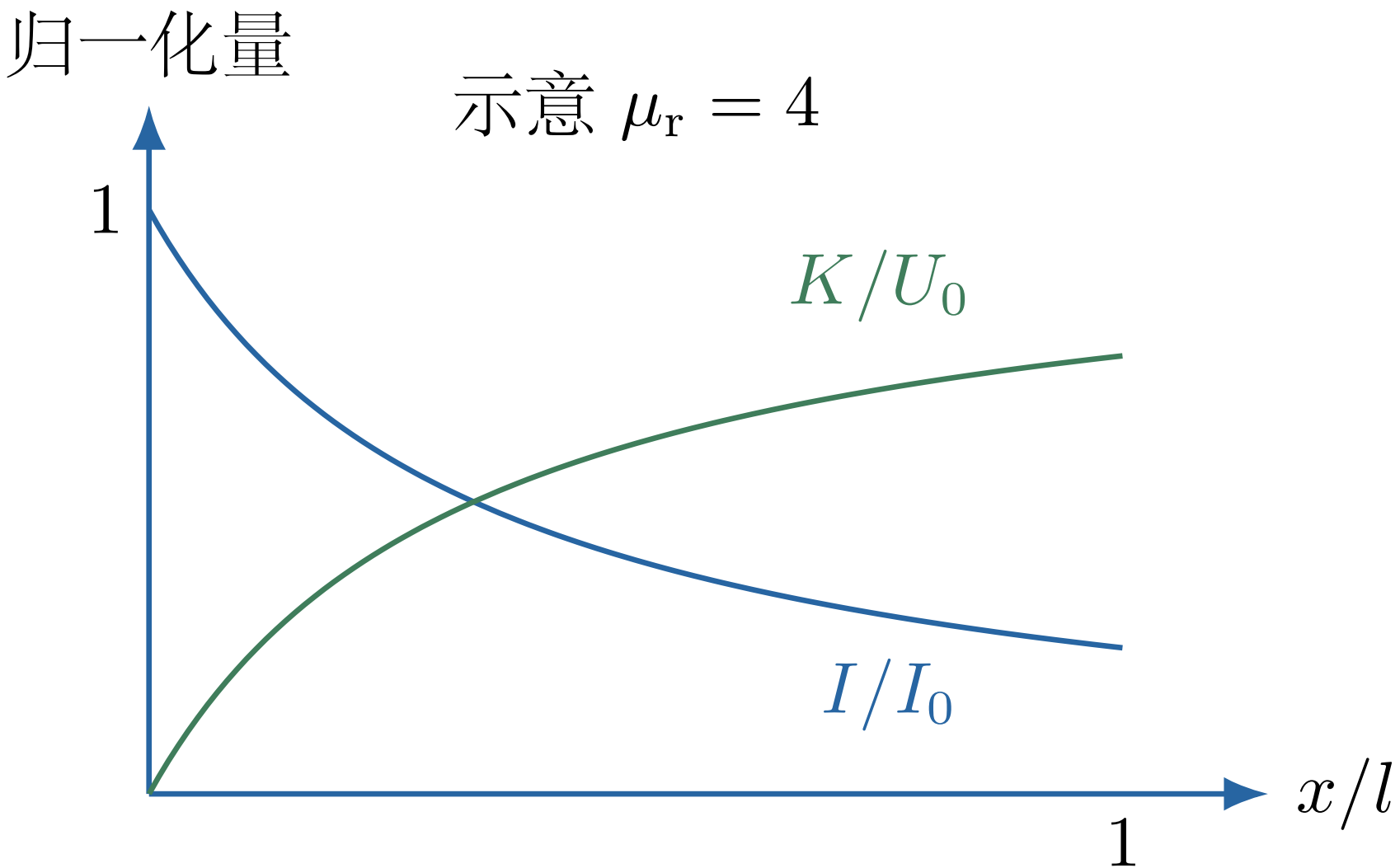

插入方向
x
I
铁芯 μ_{r}
r
B
l



所围匝数 na


$l - x$
x
F
B_{air}
B_{core}
插入为正

恒流电源
$\mathrm{d}W_{\mathrm{src}} = I^2\mathrm{d}L$
场能增加
$\frac{1}{2}I^2\mathrm{d}L$
机械功
$\frac{1}{2}I^2\mathrm{d}L$
无电源：Ψ 恒定
$U = \Psi^2/(2L)$
场能减少变为机械功
$-\mathrm{d}U = F\,\mathrm{d}x$


归一化量
示意 $\mu_{\mathrm{r}}=4$
1
K/U_0
I/I_0
x/l
1